

# SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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## ABSTRACT

Reliable LLM fingerprinting should support provenance verification and model attribution across the full model lifecycle, not only after downstream fine-tuning has produced mature behavioral signatures. Existing methods are largely evaluated in that late-stage setting, even though most model capacity and knowledge are acquired during pretraining. This leaves a central gap for reviewers and practitioners: can a fingerprint verify lineage from a model’s “birth” at initialization through large-scale training and later adaptation?

We answer this question by first showing that existing fingerprinting methods become *unreliable* in early-pretraining regimes because their signatures are post-hoc artifacts of training data, optimization dynamics, or hyperparameters. We then introduce **SeedPrints**, an intrinsic fingerprinting method that uses random-initialization biases as persistent, seed-dependent identifiers that exist before training begins. Our core empirical contribution is to show that untrained models already exhibit reproducible prediction biases induced by the initialization seed, and that these weak signals remain statistically detectable throughout training.

Across LLaMA-style and Qwen-style models, **SeedPrints** distinguishes models at the seed level and enables birth-to-lifecycle identity verification. Unlike prior techniques that fail during early pretraining or degrade under distribution shifts, SeedPrints remains effective from initialization to large-scale pretraining and downstream adaptation. Evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks further demonstrate robustness under prolonged training, domain shifts, and parameter modifications.

Together, these results establish initialization as a source of unique and persistent LLM identity: a true “Galtonian” fingerprint that supports lineage verification before, during, and after training. Code will be released upon acceptance.

## 1 INTRODUCTION

LLM fingerprinting is meant to give model owners and auditors a practical way to identify, attribute, and trace models from observable behavior (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). The contribution target of this paper is therefore lineage verification: establishing a verifiable link between a suspicious model and a putative source model, especially when theft or unauthorized reuse is suspected (Yoon et al., 2025; Zhang, 2025). We argue that this goal requires fingerprints that remain visible throughout the model lifecycle, including the pretraining stages where most model capacity is formed.

The fingerprinting literature often invokes Francis **Galton**’s biological metaphor from *Finger Prints* (1892) (Galton, 1892):

*“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”*

Taken seriously, this metaphor defines a stronger reviewer-facing standard for LLM fingerprinting: a fingerprint should be unique, persistent, and present at the model’s “birth” at initialization. Existing

## SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

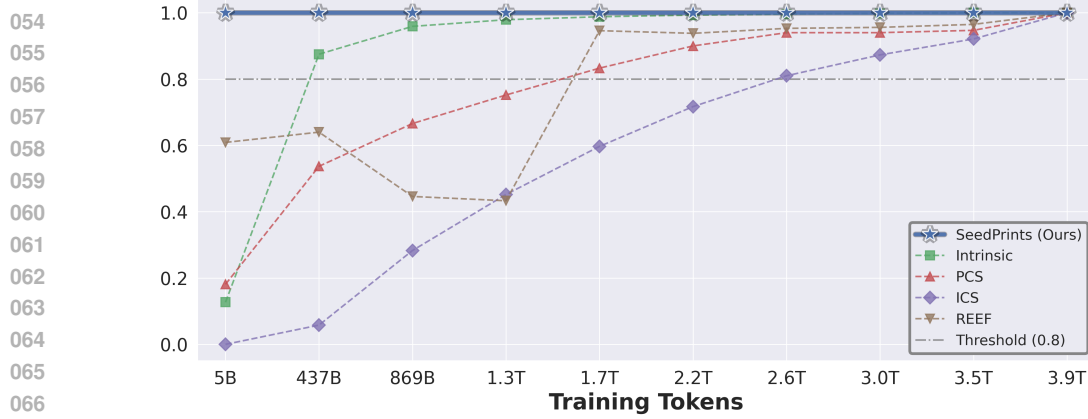


Figure 1: **SeedPrints closes the early-pretraining gap left by existing fingerprinting methods.** We compare five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint. The y-axis reports the lineage score (higher indicates a stronger lineage signal); the dashed line marks the 0.8 detection threshold used by similarity-based baselines. All baselines degrade and fall below the threshold at early checkpoints, whereas **SeedPrints** achieves perfect detection ( $p \ll 0.001$ , plotted as  $1 - p$ ) from the first checkpoint onward.

approaches generally do not satisfy this standard (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As shown in Figure 1, they fail to verify lineage reliably during early pretraining. The reason is structural: their signatures are typically extracted only after a model has converged, for example from parameters or generated text. Consequently, their separability often reflects training-induced surface factors—data signatures, optimization dynamics, or hyperparameters—rather than an intrinsic birthmark. In Section 5.1, we further show that these methods can fail when the training distribution changes substantially. Thus, they operate more like post-hoc identifiers than **Galtonian fingerprints**: innate and persistent marks that accompany a model from the beginning.

This paper contributes a stricter notion of LLM fingerprinting: an intrinsic property that is already present at *initialization* and remains detectable at *any* later training stage. We instantiate this notion with **SeedPrints**. The contributions are deliberately organized around the questions a lineage-verification method must answer:

- **We identify a fundamental evaluation gap in existing fingerprints.** Prior methods fail to verify lineage during early pretraining and can be misled by large shifts in training-data distribution, showing that late-stage fine-tuning evaluations do not establish lifecycle identity. (Section 5.1)
- **We uncover initialization-driven fingerprint signals.** Untrained models exhibit seed-dependent biases in their internal representations, creating a weak but persistent identity signal from initialization. (Section 3)
- **We propose SeedPrints as a statistical lineage test.** SeedPrints isolates initialization-born bias through identity dimensions and tests whether two models agree more than expected under a calibrated null, reducing reliance on heuristic similarity thresholds and on confounds such as data distribution, training duration, and optimization dynamics. (Section 4)
- **We validate birth-to-lifecycle verification.** SeedPrints distinguishes models that differ only in initialization seed, remains stable when identical data and order are used, persists under continual training on diverse datasets, and continues to verify lineage across large-scale pretraining, downstream adaptation, and practical deployment scenarios on *LeaFBench* (Shao et al., 2025). (Sections 5.1 to 5.3)

The resulting message is simple: initialization itself can imprint a persistent identity on LLMs. By making this signal measurable, SeedPrints moves fingerprinting from post-hoc model recognition toward lifecycle provenance verification.

## 2 RELATED WORK

Prior LLM protection work can be organized around whether the signature is inserted or discovered. This distinction is important for our contribution because SeedPrints is designed to discover a pre-existing, initialization-born signal rather than to modify a model. The two main families are (i)

watermarking / active fingerprinting methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s existing behavior without changing it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

**Watermarking and Active Fingerprinting.** Active methods contribute verifiable ownership evidence by deliberately implanting identifiers for later checks. Classic examples include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures, and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, the common variants are *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), and *model weight watermarks*, which embed identifiers into parameters or bind them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). These approaches can be effective when the owner controls training, and backdoor-style fingerprints may persist after moderate fine-tuning (e.g., Dasgupta et al. 2024). Their limitation for our setting is equally clear: because they are invasive, they cannot retroactively identify lineage in third-party models that were not marked in advance.

**Passive LLM Fingerprinting.** Passive fingerprinting is closer to our goal because it identifies models by analyzing intrinsic properties without modifying them. Existing passive methods differ mainly by access regime. With **white-box access**, signatures are extracted from model weights or representations, such as attention-matrix statistics (Yoon et al., 2025), kernel alignment of internal representations (Zhang et al., 2024), or stable parameter-vector directions. In the **black-box setting**, methods instead rely on input-output behavior, including crafted queries (Pasquini et al., 2024), distinctive prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024), though such signals can be less robust to fine-tuning. The contribution boundary of SeedPrints is different: existing passive methods define signatures *post hoc*, after training has produced identifiable properties, whereas we seek the innate, “Galtonian” signal already present at random initialization. This distinction explains why prior methods are not designed for large-scale pretraining lineage verification, where late-stage post-training signatures may not yet exist.

### 3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

This section provides the empirical observation that makes SeedPrints possible: random initialization leaves seed-specific prediction biases, and training weakens but does not erase them. The contribution of this section is not a new training procedure, but evidence that a usable fingerprint signal exists before any data is seen.

**Initialization induces seed-specific bias profiles.** We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, where tokens are sampled uniformly from the vocabulary. In Figure 2 (*Left*), we measure how often each output token (top, final logits) and each hidden dimension (bottom, last-layer representations) attains the minimum value across these random trials. The resulting distributions are sharply non-uniform, showing that even an untrained model has extreme output-preference biases. Repeating the same experiment across different initialization seeds reveals the fingerprinting opportunity: the magnitude of the bias is stable, but the specific argmin dimensions are seed-dependent (Figure 2, *Lower Right*). These observations align with recent evidence that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and amplified by self-attention (Li et al., 2026). Under this interpretation, different seeds produce different contraction directions and therefore different argmin patterns across output dimensions.

**Training preserves initialization-born bias.** The next question is whether this seed-specific birth signal survives training strongly enough to support lineage verification. To test this, we train the same seed-123 initialization for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints. We focus on the  $m = 50$  output dimensions that the initialized model most frequently assigns minimum values. For each checkpoint, we compute the correlation between its responses and the initialized model’s responses across random inputs, restricted to these dimensions; this statistic measures whether the checkpoint and its initialization rank or disfavor the same dimensions in similar ways. As a negative baseline, we compute the same correlation against independently initialized models (seed 1000). As shown in Figure 2 (*Upper Right*), same-lineage correlations are

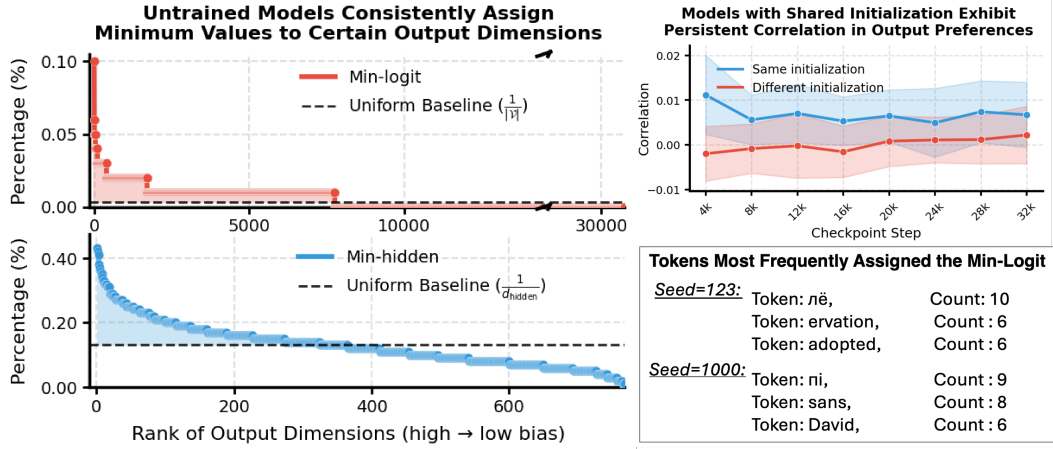


Figure 2: **The empirical basis for SeedPrints: initialization-born output bias is seed-specific and training-persistent.** *Left:* Under completely random inputs, a randomly initialized LLaMA-2-style model produces non-uniform outputs: some dimensions are repeatedly disfavored, i.e., they frequently receive the minimum value across random inputs. This extreme bias appears in both logits (top, red) and final hidden representations (bottom, blue). The dashed line marks the expected frequency under a uniform distribution, and the arrows in the top panel indicate a broken x-axis that omits low-frequency tail ranks. *Upper Right:* During training, checkpoints remain weakly but consistently correlated with their own initialization on these bias patterns. The same-lineage distribution is shifted upward relative to correlations with independently initialized models, showing that the initialization signal remains measurable. *Lower Right:* The identity-bearing dimensions are seed-dependent; the panel illustrates this using tokens that most frequently receive the minimum logit.

small in absolute value but consistently shifted upward relative to the independent baseline, and they stabilize rather than decaying to zero. This establishes the key signal used by SeedPrints: initialization leaves a weak but statistically detectable bias that persists throughout training.

#### 4 ALGORITHM

SeedPrints turns the observation in Section 3 into a practical lineage-verification test. Given two models, we do not assume access to the original initialized model. Instead, the method (i) extracts candidate identity dimensions from each model, (ii) measures their agreement across random inputs, and (iii) asks whether that agreement is significantly larger than expected between unrelated models. The full procedure is summarized in Algorithm 1.

**Contribution 1: extracting identity dimensions between any two models.** Let  $X = \{x_i\}_{i=1}^n$ , where each  $x_i \in \mathbb{R}^{\ell \times d}$  denotes a random sequence of length  $\ell$ . Each random input  $x_i$  can be instantiated either by uniformly sampling  $\ell$  token IDs from the vocabulary, or by directly sampling  $\ell$  independent vectors from a  $d$ -dimensional isotropic Gaussian distribution. For any model  $g$ , we define its mean response vector as  $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$ , where  $d_{\text{out}}$  denotes the output dimensionality. Here,  $g(x_i)$  can denote either the model’s logits or its final-layer representations. We then identify its disfavored dimensions as the set of bottom- $m$  coordinates of  $\bar{g}$ :

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

The mean response vector is used because it is a stable and noise-robust way to recover the bias profile, compared with per-sample argmin frequency (i.e., counting how often each dimension attains the minimum value, as in Figure 2). Appendix C.1 provides the formal comparison.

Given two models  $f$  and  $f'$ , we define their *identity dimensions* as the intersection  $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$ . This intersection is the first reviewer-facing design choice in SeedPrints: if two models share an initialization lineage, they should independently recover overlapping disfavored dimensions. By requiring mutual agreement, the method suppresses correlations caused only by architecture or shared training dynamics and concentrates on signals more likely to originate from the shared seed. When one model is close to initialization, its disfavored dimensions can also serve directly as an anchor.

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**Algorithm 1** Distribution Correlation Test on Identity Dimensions

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**Require:** base model  $f$ , suspicious model  $f'$ ; random inputs  $X = \{x_i\}_{i=1}^n$ ; fingerprint size  $m$ ; significance level  $\alpha$

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs  $\bar{f}, \bar{f}'$  over  $X$
- 2:  $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$  bottom- $m$  dimensions of  $\bar{f}, \bar{f}'$
- 3:  $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$  (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize  $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$  for each  $x_i \in X$  across identity dimension
- 5: **for** each  $s_j \in \mathcal{S}$  **do**
- 6:    $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$  (correlation in in input-wise preferences)
- 7: **end for**
- 8:  $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{T}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct  $\mathcal{T}_{\text{null}}$  by applying the same pipeline to two Gaussian matrices  $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test  $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$  vs.  $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage  $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

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**Contribution 2: measuring agreement as a correlation distribution.** After restricting both models to the identity dimensions  $\mathcal{S}$ , SeedPrints compares their responses across random inputs.<sup>1</sup> For each  $j \in \mathcal{S}$ , we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This produces  $T = \{\tau_j\}_{j \in \mathcal{S}}$ , a distribution of agreement scores across identity dimensions. The contribution of using a distribution rather than a single score is that the test can aggregate many weak seed-born signals: as observed in Section 3, same-lineage models have distributionally higher agreement on these dimensions than unrelated models.

**Contribution 3: calibrated hypothesis testing against a null.** Because same-lineage and different-lineage correlation distributions can overlap, SeedPrints does not rely on a heuristic threshold for an individual statistic. Instead, it tests whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. The ideal null would be estimated from correlations between independent models, as in Figure 2, but doing so requires many independently initialized models and is computationally expensive. Appendix B shows that this independent-model null is well approximated by a Gaussian distribution, motivating a cheaper simulation-based proxy that passes Gaussian surrogates through the same pipeline, as summarized in Algorithm 1.

The simulation-based null is useful, but it adds randomness and overhead: its precision depends on the number of trials, results may vary across runs, and comparing two empirical distributions introduces test-specific assumptions such as those used by the one-sided t-test or Mann–Whitney U test. SeedPrints therefore also provides an analytical test based on the closed-form null distribution of Kendall–Tau under weak independence, while retaining both empirical and analytical implementations in the code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance  $\sigma^2$ . By the central limit theorem, as the number of identity dimensions  $|\mathcal{S}|$  increases, the distribution of  $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$  is well approximated by

$\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$ . Although softmax normalization introduces weak dependencies across dimensions, the high temperature ( $T = 10$ ) mitigates these effects, making the independence approximation practical. We empirically validate that this Gaussian approximation closely matches the full-pipeline distribution in Appendix B. We then compute the one-sided z-score  $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$  and the corresponding

p-value  $p = 1 - \Phi(z)$ , where  $\Phi$  is the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage if  $p < 0.01$ .

**Practical instantiation.** SeedPrints can use either logits or hidden representations as model outputs. Both choices match the formulation, but logits require many more random inputs as vocabulary size grows, making reliable bias estimation substantially more expensive (see Appendix C.2 for the theoretical analysis). For efficiency and stability, our default implementation uses final-layer hidden

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<sup>1</sup>Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Table 1: SeedPrints separates initialization seeds yield non-shared fingerprint behavior. Table 2: SeedPrints verifies inheritance from birth to pretrain-models at birth: different initialization seeds yield non-shared fingerprint behavior: each trained model matches its own initialization ( $p$ -value  $< 0.01$ ).

| Model Pairs               | $p$ -value ( $> 0.01$ ) | Model Pairs                             | SeedPrints | Baselines |        |        |        |
|---------------------------|-------------------------|-----------------------------------------|------------|-----------|--------|--------|--------|
|                           |                         |                                         | $p$ -value | Intrinsic | REEF   | PCS    | ICS    |
| $s_{42}$ VS. $s_{2000}$   | 0.116                   | $s_{42}^{init}$ vs. $s_{42}^{base}$     | 3.37e-26✓  | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s_{123}$ VS. $s_{42}$    | 1.000                   | $s_{123}^{init}$ vs. $s_{123}^{base}$   | 3.16e-33✓  | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s_{1000}$ VS. $s_{123}$  | 0.327                   | $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 1.12e-21✓  | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s_{2000}$ VS. $s_{1000}$ | 1.000                   | $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 3.70e-31✓  | -0.337×   | 0.331× | 0.581× | 0.188× |

states. The method’s output is a  $p$ -value, giving a direct statistical lineage test rather than a heuristic similarity score.

## 5 EXPERIMENTS

The experiments are organized to make the contribution claims directly testable. First, in Section 5.1, we ask whether SeedPrints behaves like a genuine fingerprint: (i) it supports **birth verification at the seed level**, and (ii) it remains **verifiable across the entire training lifecycle**. This setting also exposes the central limitation of existing baselines: they do not verify lineage during early pretraining and often rely on data-dependent signals that fail under large distribution shifts. Unless otherwise stated, we report SeedPrints with the analytical null and *hidden state* outputs; Appendix D.2 reports complementary empirical-null results using both *logit* and *hidden state* outputs with the one-sided  $t$ -test and Mann–Whitney  $U$  test. Second, in Section 5.3, we test whether the same contribution remains useful in practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Across 65 model instances and 6 representative post-development techniques, SeedPrints matches the strongest baselines in post-training settings while preserving its distinctive lifecycle-verification advantage.

**Baselines.** We compare against four passive fingerprinting baselines that represent weight- and representation-based attribution. **Intrinsic fingerprint** (Yoon et al., 2025) (also called *PDF* in some papers) compares layerwise standard-deviation profiles of attention parameters. **REEF** (Zhang et al., 2024) measures centered-kernel-alignment (CKA) similarity between feature representations from the same samples across two models. **PCS** and **ICS** (Zeng et al., 2024) (collectively called *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and computes cosine similarity, whereas ICS forms invariant terms from the weights and computes cosine similarity on those invariants. In Section 5.3, we also include the weaker gradient-based fingerprint **Gradient** (Shao et al., 2025). Following Zhang et al. (2024), similarity-based baselines use a **0.8 similarity threshold** for binary decisions.

For scanability, all experiment tables use color to mark the ground-truth lineage relation: **green** denotes models from the same source, and **red** denotes different sources. For example,  $s_{42}^{init}$  vs.  $s_{42}^{base}$  compares a seed-42 initialization with its continued-pretrained descendant, whereas  $s_{2000}^{init}$  vs.  $s_{42}^{base}$  compares a seed-2000 initialization with a model trained from seed 42. A ✓ indicates a correct detection, and × indicates an error.

### 5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

This subsection validates the core contribution under controlled training: the fingerprint is seed-specific at initialization and remains detectable after training. We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports LLaMA-style results, while Qwen-style results are deferred to Appendix D; both model families support the same conclusions.

**Different initialization seeds produce distinct fingerprints.** Table 1 tests the first requirement of a birth fingerprint: different seeds should not be confused. We report  $p$ -values from SeedPrints correlations between models initialized with seeds 42, 123, 1000, and 2000. All  $p$ -values remain  $> 0.01$ , so the method does not declare shared lineage across different seeds. This result establishes seed-level distinguishability before training can introduce data- or optimization-dependent artifacts.

Table 3: SeedPrints is not a data-order artifact: identical corpus and curriculum do not make different initializations share fingerprints.

| Model Pair                              | $p$ -value ( $> 0.01$ ) |
|-----------------------------------------|-------------------------|
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.573                   |
| $s_{123}^{init}$ vs. $s_{2000}^{base}$  | 0.724                   |
| $s_{2000}^{init}$ vs. $s_{1000}^{base}$ | 0.109                   |
| $s_{1000}^{init}$ vs. $s_{42}^{base}$   | 0.883                   |

Table 4: SeedPrints preserves lineage under domain shift during continual training (base model: seed 1000, corpus openwebtext).

| Setting                 | SeedPrints    | Baselines |        |        |        |
|-------------------------|---------------|-----------|--------|--------|--------|
| Continual corpus (seed) | $p$ -value    | Intrinsic | REEF   | PCS    | ICS    |
| TinyStories (1000)      | $\approx 0$ ✓ | 1.000✓    | 0.759× | 0.999✓ | 0.996✓ |
| TinyStories (123)       | 1.000✓        | 0.950×    | 0.658✓ | 0.332✓ | 0.012✓ |
| the_stack (1000)        | $\approx 0$ ✓ | 0.489×    | 0.557× | 0.585× | 0.123× |
| the_stack (123)         | 1.000✓        | 0.445✓    | 0.580✓ | 0.301✓ | 0.026✓ |

**Training preserves the initialization fingerprint.** Table 2 tests the second requirement: a trained descendant should retain the fingerprint of its initialization. We compare each initialization model  $s^{init}$  with its corresponding  $s^{base}$  model trained on OpenWebText (Gokaslan et al., 2019) ( $\approx 10B$  tokens). For every seed, SeedPrints returns  $p < 0.01$ , indicating that the trained model and its initialization share a statistically detectable bias profile. In contrast, the baseline methods all fail to distinguish across seeds in this setting, suggesting that their separability is dominated by training-induced artifacts rather than initialization-born identity.

**Identical data and order do not make fingerprints converge.** Table 3 isolates initialization from data order. All four suspicious models  $s_i^{base}$  for  $i \in \{42, 123, 100, 2000\}$  are trained on *exactly the same corpus (OpenWebText) and in the same data order*; we fix the training seed to lock the curriculum, so the only difference is the initialization seed  $i$ . Across all cross-seed pairs,  $p$ -values remain  $> 0.01$ , contrasting with the near-zero same-seed values in Table 2. Thus, SeedPrints does not merely recover shared data or curriculum: the fingerprint remains seed-specific even when the training pipeline is otherwise identical.

**Continual training on diverse datasets does not confound the fingerprint.** The preceding experiments establish SeedPrints as a biometric-style fingerprint, but the copyright-relevant challenge is different: attribution should not be misled by *data distribution*. Prior baselines’ inability to separate different seeds is not itself a copyright failure, because initialization seeds have no clear copyright status. Their practical fragility is that distribution shifts during continued training can make true descendants appear unrelated.

In Table 4, we continue training a seed-1000 base model pretrained on OpenWebText (Gokaslan et al., 2019) on two divergent corpora: TinyStories (Eldan & Li, 2023), consisting of synthetic children’s stories, and The Stack (Kocetkov et al., 2022), consisting of permissively licensed GitHub code. We compare true descendants trained from this base against *distractors* derived from a different seed-123 base. Because the baseline methods are insensitive to initialization differences, the distractor base is pretrained with a different OpenWebText data order and then continually trained on the same target corpus. The evaluation question is therefore whether a method identifies the descendant that actually shares lineage with the source, rather than the model with the closest downstream data distribution.

The baselines all fail in the code-domain setting (The Stack), misclassifying true descendants as distractors. This pattern indicates that they largely track domain similarity: TinyStories is closer to OpenWebText, whereas The Stack is a sharper distribution shift. SeedPrints correctly attributes lineage across both corpora, showing that its fingerprint is not simply a proxy for data distribution and remains detectable after substantial continued training.

## 5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** We next ask whether SeedPrints can verify offspring throughout a training trajectory, not only at initialization and final checkpoints. In a controlled OpenWebText pretraining run with known initialization seed, we test intermediate checkpoints and observe consistent detection at all stages (Appendix D.2.1, Figure 5). This controlled result supports the lifecycle claim under a setting where lineage is fully known.

**Large-scale pretraining.** To test the same claim under realistic foundation-model training, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B  $\rightarrow$  3.9T tokens). We treat the final checkpoint as the target model and each earlier checkpoint as a tested ancestor. This setting is harder than the controlled OpenWebText run because it includes heterogeneous corpora,

Table 5: SeedPrints remains verifiable after large-scale finetuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the  $p$ -value from our correlation test ( $< 0.01$  indicates a strong signal). Four baselines all report similarity scores (threshold = 0.8, higher = better).

| Model                               | # Tokens | SeedPrints $p$   | Intrinsic $\uparrow$ | REEF ( $\uparrow$ ) | PCS ( $\uparrow$ ) | ICS ( $\uparrow$ ) |
|-------------------------------------|----------|------------------|----------------------|---------------------|--------------------|--------------------|
| Llama-2-finance-7B (Heenan, 2023)   | 5M       | $10^{-41955}$ ✓  | 1.0000✓              | 0.9950✓             | 0.9979✓            | 0.9952✓            |
| Vicuna-1.5-7B (Chiang et al., 2023) | 370M     | $10^{-103043}$ ✓ | 1.0000✓              | 0.9985✓             | 0.9985✓            | 0.9949✓            |
| Wizardmath-7B (Luo et al., 2023)    | 1.8B     | $10^{-180550}$ ✓ | 1.0000✓              | 0.9979✓             | 1.0000✓            | 0.9994✓            |
| Meditron-7B (Chen et al., 2023)     | 48B      | $10^{-42566}$ ✓  | 0.9990✓              | 0.9978✓             | 1.0000✓            | 0.9817✓            |
| CodeLlama-7B (Meta AI, 2024)        | 500B     | $10^{-3552}$ ✓   | 0.9480✓              | 0.9947✓             | 0.6863✗            | 0.3369✗            |
| Llemma-7B (Azerbayev et al., 2023)  | 700B     | $10^{-5136}$ ✓   | 0.9470✗              | 0.9984✓             | 0.6682✗            | 0.2905✗            |

longer optimization, and large-scale representation drift. For comparability with similarity-based baselines, we plot  $1 - p$ ; all SeedPrints  $p$ -values are  $\ll 0.01$  and numerically close to zero.

Figure 1 highlights the main experimental contribution: **existing baselines fail during early pretraining** (e.g.,  $ICS \approx 0$ ,  $PCS < 0.3$ , and  $REEF < 0.6$  within the first **trillion** tokens), whereas SeedPrints remains verifiable and strengthens throughout training. This exposes a gap in prior evaluation practice: most work verifies lineage after fine-tuning, where detection is easier and can overstate safety. Our results show that pretraining itself can reshape representations and produce stronger later-stage signals, so lineage verification should prioritize early pretraining, when misuse is hardest to detect.

**Large-scale finetuning.** We also evaluate whether SeedPrints retains utility under standard fine-tuning benchmarks. Suspect models are fine-tuned from Llama-2-7b with data volumes ranging from 5 million to 700 billion tokens, including Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), CodeLlama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023). Their fine-tuning data volumes are 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As shown in Table 5, SeedPrints consistently maintains  $p < 0.01$ , supporting the claim that the initialization signal remains detectable even after large downstream updates.

### 5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real deployments often alter foundation models through fine-tuning, PEFT, quantization, merging, and distillation. This subsection asks whether the SeedPrints contribution remains practically useful under such parameter-altering transformations. We use *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing, to evaluate lineage detection under realistic post-development conditions.

**Across Source Models.** We audit two source types: *Pre-Trained (PT)* models and *Instruction-Tuned (IT)* models. For each source type, the task is to distinguish *derivative* models, created by post-training adaptations, from *independent* models that do not use the source weights. This mirrors practical auditing, where the source may be either a PT base or an IT variant. The evaluation covers **65** models; Appendix E lists all model names and HuggingFace repositories.

**Metrics.** We report two measures of detection performance: (i) Area Under the ROC Curve (AUC), and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which measures the maximum separation between derivative and independent score distributions.

SeedPrints outputs  $p$ -values, whereas the baselines output similarity scores. This distinction matters for interpreting the contribution: similarity scores are linear measures, but  $p$ -values are tail probabilities under a null hypothesis and should not be read as linear similarity. To compute AUC in the same pipeline as the baselines, we convert  $p$ -values to scores using  $s = 1 - p$ . This conversion is convenient but not perfectly aligned with the statistical meaning of  $p$ -values, because small  $p$ -values vary on an exponential scale while threshold sweeping is linear. Even under this conservative comparison, SeedPrints achieves comparable or superior performance. We compute AUC by sweeping thresholds over  $s$  for all methods and report KS as a threshold-free separability measure. We do not report Manhattan distance because absolute distances between  $p$ -values are not meaningful.

In practical audits, calibrating an optimal similarity threshold is often infeasible. SeedPrints contributes a different decision interface: for any model pair, it returns a  $p$ -value for the shared-lineage

Table 6: Family-wise deployment robustness of LLM fingerprinting methods. “PT” and “IT” use the pre-trained and instruction-tuned models as source models, respectively. Per-family results cover the 6 families with both PT and IT models; “Overall” scores are computed over all test model pairs rather than averaged across families.

| Method     | Metric                  | Qwen2.5-7B |       | Qwen2.5-14B |       | Llama3.1-8B |       | Mistral-7B |       | Gemma2-2B |       | Llama2-7B |       | Overall      |
|------------|-------------------------|------------|-------|-------------|-------|-------------|-------|------------|-------|-----------|-------|-----------|-------|--------------|
|            |                         | PT         | IT    | PT          | IT    | PT          | IT    | PT         | IT    | PT        | IT    | PT        | IT    |              |
| REEF       | AUC $\uparrow$          | 0.796      | 0.862 | 0.947       | 0.913 | 1.000       | 0.999 | 0.997      | 0.997 | 0.963     | 0.968 | 0.750     | 0.706 | <b>0.915</b> |
|            | KS Statistic $\uparrow$ | 0.492      | 0.633 | 0.875       | 0.735 | 0.987       | 0.987 | 0.975      | 0.980 | 0.950     | 0.837 | 0.750     | 0.694 | <b>0.739</b> |
| Gradient   | AUC $\uparrow$          | 0.657      | 0.743 | 0.763       | 0.789 | 0.897       | 0.904 | 0.872      | 0.682 | 1.000     | 1.000 | 0.650     | 0.768 | <b>0.801</b> |
|            | KS Statistic $\uparrow$ | 0.354      | 0.423 | 0.500       | 0.563 | 0.646       | 0.705 | 0.725      | 0.500 | 1.000     | 1.000 | 0.275     | 0.479 | <b>0.508</b> |
| ICS        | AUC $\uparrow$          | 1.000      | 1.000 | 0.997       | 0.997 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic $\uparrow$ | 1.000      | 1.000 | 0.975       | 0.979 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.943</b> |
| Intrinsic  | AUC $\uparrow$          | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic $\uparrow$ | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.989</b> |
| SeedPrints | AUC $\uparrow$          | 0.992      | 0.994 | 0.988       | 0.990 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.992</b> |
|            | KS Statistic $\uparrow$ | 0.985      | 0.987 | 0.975       | 0.980 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.986</b> |

hypothesis. **This enables a direct statistical decision without tuning a similarity threshold**, while the AUC and KS results below are included for benchmark comparability.

**Parameter-Altering Techniques.** We evaluate robustness across six post-development transformations: (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

### 5.3.1 FAMILY-WISE EFFECTIVENESS

There are seven model families in total. For each family, we treat its pretrained base model (PT) and instruction-tuned model (IT) as separate sources and compare them with all other models in the 65-model pool. Table 6 reports the results; TinyLlama-1.1B families are omitted from Table 6 because they are trivial for all methods. Each family column reports AUC over tests for that family using a family-calibrated threshold, while the overall score is computed across all test pairs using a global threshold; therefore, the overall score is not the average of the family-wise scores. The key result is that *Intrinsic*, *ICS*, and *SeedPrints* are all essentially saturated across alterations, with overall AUCs of 0.994, 0.994, and 0.992, and KS scores of 0.989, 0.943, and 0.986, respectively. In most families these methods obtain full scores, whereas *REEF* and *Gradient* perform substantially worse. Thus, although *SeedPrints* is designed around intrinsic seed-level identity, it also functions as a reliable biometric-like fingerprint across diverse deployed model families.

### 5.3.2 ALTERATION-WISE ROBUSTNESS

The alteration-wise results clarify where the contribution is strongest and where the initialization signal can be diluted. *Instruct* is consistently the easiest case, with even the weakest baseline, *Gradient*, reaching AUC 0.895. *Finetune* and *Merge* are more challenging across methods; for *Finetune*, no method is perfect. *PEFT*, *Quantization*, and *Distillation* show mixed effects: *REEF* drops under *Quantization* ( $0.996 \rightarrow 0.871$ ) and *Distillation* ( $0.996 \rightarrow 0.858$ ) while staying resilient to *PEFT* (0.994), whereas *PEFT* and *Quantization* are harder for *Gradient* ( $0.895 \rightarrow 0.776/0.756$ ). The three stronger methods remain mostly robust, with all scores  $> 0.9$ . *SeedPrints* drops slightly to 0.959 on *Merge*, failing on one merge model out of seventeen; this suggests that weight-space interpolation with unrelated weights can dilute the initialization signal below the detection threshold. Averaged across all six alteration types, *Intrinsic* has the highest overall AUC, followed closely by *ICS* and *SeedPrints*.

## 5.4 ABLATION STUDY ON HYPERPARAMETERS

The ablation studies make the method’s practical tradeoffs explicit. *SeedPrints* has three hyperparameters: the number of random input sequences  $n$ , the length of each random sequence  $l$ , and the fingerprint size  $m$ . The first two control the amount of random-input evidence, while  $m$  controls how many low-mean coordinates enter the identity-dimension test. In the main experiments in Sections 5.2 and 5.3, we set  $n=2000$ ,  $l=1024$ , and  $m=0.1d_{\text{out}}$ , where  $d_{\text{out}}$  is the hidden-state dimensionality, so the setting scales with model size. Here, we vary one hyperparameter at a time while fixing the others

Table 7: Alteration-wise robustness of LLM fingerprinting methods across parameter-changing deployment techniques.

| Method     | Metric                  | Instruct | Finetune | PEFT  | Quantization | Merge | Distillation | Overall Avg  |
|------------|-------------------------|----------|----------|-------|--------------|-------|--------------|--------------|
| REEF       | AUC $\uparrow$          | 0.996    | 0.872    | 0.994 | 0.871        | 0.954 | 0.858        | <b>0.924</b> |
|            | KS Statistic $\uparrow$ | 0.949    | 0.637    | 0.940 | 0.812        | 0.851 | 0.615        | <b>0.801</b> |
| Gradient   | AUC $\uparrow$          | 0.895    | 0.729    | 0.776 | 0.756        | 0.832 | 0.887        | <b>0.812</b> |
|            | KS Statistic $\uparrow$ | 0.756    | 0.441    | 0.469 | 0.586        | 0.519 | 0.679        | <b>0.575</b> |
| ICS        | AUC $\uparrow$          | 1.000    | 0.987    | 1.000 | 1.000        | 0.981 | 1.000        | <b>0.995</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.925    | 1.000 | 1.000        | 0.941 | 1.000        | <b>0.978</b> |
| Intrinsic  | AUC $\uparrow$          | 1.000    | 0.980    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.997</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.971    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.995</b> |
| SeedPrints | AUC $\uparrow$          | 1.000    | 0.995    | 0.990 | 1.000        | 0.959 | 1.000        | <b>0.991</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.990    | 0.980 | 1.000        | 0.941 | 1.000        | <b>0.985</b> |

to their defaults (i.e.,  $n = 2000$ ,  $m = 400$ ,  $l = 768$ ), and report empirical accuracy and false-positive rate at significance level  $\alpha = 0.01$  in Table 8.

The main ablation result is that increasing  $n$  and  $l$  improves identification performance. This matches the theory in Section C.2: to recover the population-level bias accurately, the required number of queries must satisfy  $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$ , where  $\gamma$  is the bias margin separating biased coordinates from the remaining dimensions. Smaller  $n$  therefore gives too little evidence for reliable coordinate recovery. Longer sequences increase  $\gamma$  through accumulation effects, strengthening separation; under a fixed and potentially insufficient number of sequences, increasing  $l$  reduces the query burden and improves detection.

The fingerprint size  $m$  shows the expected non-monotonic tradeoff. If  $m$  is too small, the selected coordinates are unstable because there are too few observations to form a reliable correlation distribution. If  $m$  is too large, many irrelevant coordinates enter the test, shrinking  $\gamma$  and requiring quadratically more samples to compensate. Empirically,  $m$  in the range 200–400 works well across model sizes, consistent with the heuristic  $m \approx 0.1d_{\text{out}}$  for typical  $d_{\text{out}} \in [3000, 5000]$ .

## 6 CONCLUSION

This paper makes the contribution structure of LLM fingerprinting more stringent: a useful fingerprint should support lineage verification from initialization through training and deployment. We introduced *SeedPrints*, an intrinsic fingerprinting method that traces a model’s lineage back to its random initialization. The central finding is that initialization itself leaves a persistent “Galtonian” fingerprint: untrained models already exhibit stable, seed-dependent token-preference patterns, and these biases remain statistically measurable throughout training. Building on this signal, *SeedPrints* uses a distribution-correlation test over *identity dimensions* to detect shared lineage with calibrated significance. Across LLaMA- and Qwen-style families and realistic deployment transformations—including continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation—*SeedPrints* enables birth-to-lifecycle verification and remains robust under domain shift, supporting provenance and copyright auditing.

**Limitations and Future Directions** The contribution of *SeedPrints* is currently strongest in settings where model outputs such as logits or hidden states are accessible. It may not apply under restricted API-only access, especially when responses are deterministic, heavily filtered, or otherwise insufficient for estimating hidden bias patterns. At the same time, this limitation underscores the broader research gap: most existing methods, particularly black-box methods, rely on signals that emerge only after later training or alignment and therefore do not robustly protect pretrained or

Table 8: Hyperparameter ablation for *SeedPrints* at significance level  $\alpha = 0.01$ .

| Ablation on $n$ | 200    | 500    | 2000   |
|-----------------|--------|--------|--------|
| ACC             | 0.7368 | 0.7778 | 0.9375 |
| Empirical FPR   | 0      | 0      | 0      |
| Ablation on $l$ | 512    | 768    | 1024   |
| ACC             | 0.9048 | 0.9375 | 1.0000 |
| Empirical FPR   | 0.04   | 0      | 0      |
| Ablation on $m$ | 200    | 400    | 800    |
| ACC             | 0.9375 | 0.9375 | 0.875  |
| Empirical FPR   | 0.04   | 0      | 0.1463 |

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540 early-stage models. Reliable all-stage fingerprints remain urgently needed. Appendix [A.2](#) provides  
541 the full discussion.  
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## A LLM USAGE AND LIMITATION DISCUSSIONS

### A.1 LLM USAGE

We used AI assistants, including ChatGPT and Gemini, only for writing and formatting support. Their role was limited to grammar and style checks, clearer figure and table captions, and other surface-level prose edits. For programming-related tasks, we occasionally used GitHub Copilot and Claude as coding assistants, for example to support code auto-completion and debugging hints.

### A.2 LIMITATIONS AND FUTURE DIRECTIONS

SeedPrints contributes an initialization-rooted and training-persistent fingerprint signal, but its current form remains a white-box method. It may not apply when auditors have only restricted API access, especially if responses are deterministic, safety-filtered, or otherwise insufficient for estimating logits or hidden-state patterns. This limitation does not weaken the central contribution; rather, it clarifies the next research frontier. Even in white-box settings, the field is still early in developing *true fingerprints* for LLMs. Most existing approaches rely on training-induced signatures or alignment behaviors and therefore cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable across age and environment, an LLM fingerprint should persist across continued training and deployment instead of reflecting incidental artifacts. Achieving this standard under black-box access, where a substantial performance gap remains relative to white-box methods, is an important open direction. We hope SeedPrints motivates more principled and robust fingerprinting frameworks for large models.

## B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The statistical contribution of SeedPrints depends on a reliable null: the test in Section 4 declares shared lineage only when the observed correlation statistics over random inputs are unlikely under a no-lineage model. This appendix validates that null distribution, which represents correlations between two *independent models evaluated on random inputs*.

We support the null construction in three steps. First, we characterize the empirical true null using independently initialized models. Second, we show that Gaussian surrogate outputs provide a tractable simulation-based proxy for that null. Third, we validate the analytical approximation derived from the known distribution of Kendall–Tau correlations.

### B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null is the correlation distribution for two independent models on random inputs. The cleanest empirical instantiation is to compare independently initialized, untrained models with different random seeds. We estimate this true null from  $\sim 2500$  such pairs and compute Kendall–Tau correlations on 10,000 random inputs, matching the setup in Section 3. Figure 3 shows the key validation result: the empirical null is well described by a Gaussian centered at zero.

### B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null from independently initialized models is too expensive for routine auditing because it requires many models and repeated evaluations. SeedPrints therefore uses a tractable simulation-based alternative: Gaussian surrogate outputs that approximate model behavior under the no-lineage null.

This proxy is justified by two properties of the SeedPrints test. First, Gaussian approximations to neural-network outputs are common in auditing and are supported by empirical and theoretical work (Carlini et al., 2022; Lee et al., 2019). Second, the SeedPrints pipeline uses relative ordering and selection operations, such as rank correlations and bottom- $m$  selection, rather than semantic input structure. Because independently initialized models have independent parameter matrices, their null outputs should not contain strong shared structure. Independent Gaussian matrices therefore provide a simple and tractable proxy for the behavior of unrelated models.

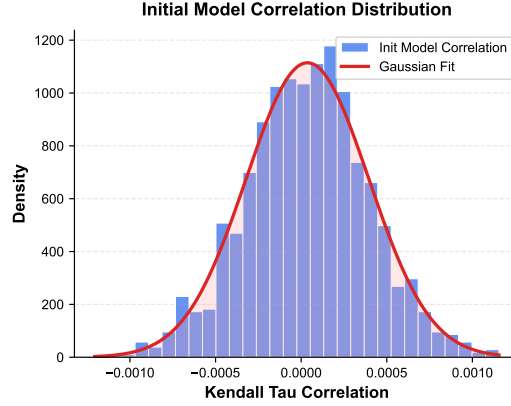


Figure 3: Empirical null validation for SeedPrints: correlation statistics over  $\sim 2500$  independently initialized model pairs, evaluated on random inputs, closely match a zero-centered Gaussian distribution.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape  $\mathbb{R}^{N \times d_{\text{out}}}$ ,
2. Extract the most biased dimensions by selecting the bottom- $m$  ranked average dimensions,
3. Take the intersection set  $\mathcal{S}$  and compute correlations on it.

This simulation-based construction applies the same selection procedure to Gaussian samples and naturally captures the dependency patterns introduced by the pipeline itself (e.g., those arising from selecting  $\mathcal{S}$  across dimensions). As shown in Figure 4 (middle), the resulting distribution matches the empirical null obtained from independently initialized models in terms of its centered Gaussian shape, indicating that the Gaussian surrogate provides an effective simulation-based approximation of the true null.

### B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

While the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it still requires repeated sampling and introduces randomness in the estimation process. We therefore also propose the most efficient analytical approximation.

Under the null hypothesis that the two models are independent<sup>2</sup>, each Kendall–Tau correlation  $\tau_j$  is approximately zero-mean with known variance  $\sigma^2$ . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. However, Figure 4 shows that the analytical Gaussian closely matches the distribution obtained from the Gaussian surrogate pipeline across different settings, including varying numbers of random inputs  $n$  and output dimensions  $D$ . This suggests that such dependencies are sufficiently weak and do not materially affect the accuracy of the approximation. A plausible explanation is that the use of a relatively high temperature ( $T = 10$ ) smooths the output distribution and reduces coupling between dimensions.

Together, these results justify the use of the analytical Gaussian null in Section 4 as an efficient, deterministic, and accurate approximation, eliminating the need for costly empirical baseline construction.

<sup>2</sup>As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

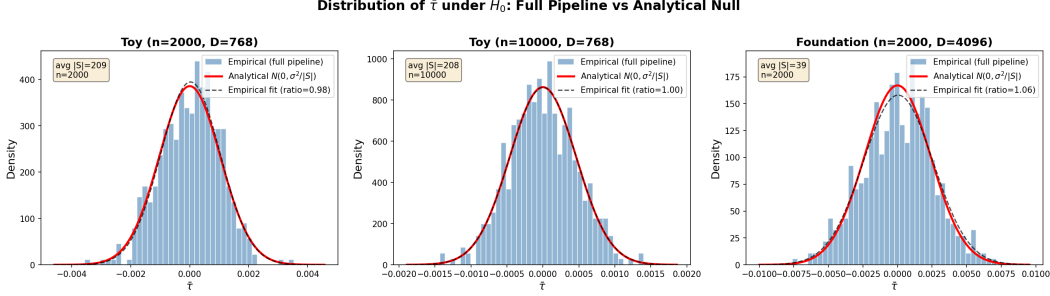


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline (including identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs and compare the resulting distribution of  $\bar{\tau}$  with the analytical Gaussian  $\mathcal{N}(0, \sigma^2/|S|)$ . The close match indicates that the analytical approximation accurately captures the null behavior, despite weak dependencies introduced by the pipeline.

**On variance scaling.** The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch does not affect our hypothesis test results. Our primary evaluation for simulation-based null in Section D.2 relies on the Mann-Whitney U-test, which is rank-based and thus invariant to any positive scaling: for any  $c > 0$ ,  $x_i < x_j \Leftrightarrow cx_i < cx_j$ . Therefore, all rank statistics and rejection thresholds remain unchanged. For tests that do depend on absolute variance, we have additionally evaluated variance-sensitive statistics such as the  $t$ -test, whose statistic scales as  $t' = t/c$  when variance decreases. Since  $c < 1$  in our case, scaling reduces the magnitude of  $t$  and makes the test more conservative rather than inflating significance. Empirically, across extensive experiments in Section D.2, we find that the variance scaling does not affect the power of the  $t$ -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, the variance is explicitly included in the normalization of the  $z$ -score. Therefore, any scale discrepancy is absorbed in the denominator. Since the analytical variance is slightly larger than the empirical one, the resulting  $z$ -scores are smaller, making the test conservative and preventing false positives.

## C THEORETICAL PROPERTIES OF OUR METHODS

Our goal is to identify the output dimensions that the initialized model is least likely to predict under random inputs. These dimensions correspond to directions that the model inherently disfavors due to initialization-induced bias (not dependent on the input distribution), forming the basis of a stable fingerprint signal.

**Setup.** Consider a model  $g$  with fixed parameters and random inputs  $x_i$  (uniform random tokens or random embeddings as in the main paper). For each output dimension  $j \in [d_{\text{out}}]$ ,  $g_j(x)$  is thus a random variable and its population mean is  $\bar{g}_j := \mathbb{E}[g_j(x)]$ . The population-level directions we aim to capture correspond to the  $m$  output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the  $m$ -th and  $(m+1)$ -th smallest means. This gap measures how well-separated the target set  $\mathcal{M}$  is from the remaining coordinates. Importantly, for a fixed  $m$  and input distribution,  $\Gamma(m)$  depends solely on the model’s inherent initialization bias. *However, increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.*

There are two natural empirical strategies to extract such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs  $X = \{x_i\}_{i=1}^n$  and select the  $m$  coordinates with the smallest value. Following the notation in Section 4, where  $g$  denotes the model (either the base model  $f$  or the suspect model  $f'$ ):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input  $x_i \in \mathbb{R}^{\ell \times d}$ , select the smallest-value coordinate and then take the  $m$  most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let  $p_j := \Pr(j^*(x) = j)$  denote the population probability that index  $j$  attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- $m$  set under voting.

Both approaches are intuitively reasonable. However, as we show below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it inherently noise-sensitive.

## C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

### C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the  $m$  smallest coordinates in the empirical mean vector, is stable under small perturbations to model outputs. The perturbations represents realistic variations in outputs from the *same* model, e.g., numerical noise, or evaluation under different precision modes.

Formally, we model perturbed outputs as  $g'(x) = g(x) + \eta(x)$ , where  $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$  represents stochastic deviations. We assume that for each sample  $x_i$  and coordinate  $j$ , the random variables  $\eta_j(x_i)$  are independent across samples (not necessarily across coordinates) and  $\sigma$ -sub-Gaussian with  $\mathbb{E}[\eta_j(x_i)] = 0$ .<sup>3</sup>

**Lemma C.1** (Stability of sample-mean-based extraction). *Let  $\hat{g}$  and  $\hat{g}'$  be the empirical means over  $X = \{x_i\}_{i=1}^n$ , i.e.,  $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$  and  $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$ . Let  $\widehat{\Gamma}(m)$  denote the empirical gap at rank  $m$ . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

**Interpretation.** The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially with the number of averaged samples  $n$ , due to variance reduction from  $\sigma^2$  per sample to  $\sigma^2/n$  in the empirical mean.

*Proof.* We condition on the sample  $X = \{x_i\}_{i=1}^n$  throughout the proof, so that  $\hat{g}$  and  $\widehat{\Gamma}(m)$  are fixed, and all probabilities are taken over the randomness of  $\{\eta(x_i)\}_{i=1}^n$  only.

<sup>3</sup>This sub-Gaussian assumption is mild: bounded random variables are  $\sigma$ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case  $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$  is a special case of this assumption.

For any coordinate  $j$ , by definition of  $\hat{g}'_j$ ,  $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$ . Each  $\eta_j(x_i)$  is mean-zero and  $\sigma$ -sub-Gaussian, so their average is mean-zero and  $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as  $1/n$  under averaging. Hence, for any  $\varepsilon > 0$ , by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than  $\varepsilon$ :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \right\}.$$

Equivalently, if  $\max_j |\hat{g}'_j - \hat{g}_j| < \widehat{\Gamma}(m)/2$ , then the bottom- $m$  identity set remains unchanged.

Let  $j_{(1)}, \dots, j_{(d_{\text{out}})}$  be a permutation of  $\{1, \dots, d_{\text{out}}\}$  such that  $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$ . Recall that the empirical gap at rank  $m$  is  $\widehat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$ . Denote  $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$  and  $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$ . For any  $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$  and any  $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$ ,  $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \widehat{\Gamma}(m)$ . With  $|\hat{g}'_i - \hat{g}_i| < \widehat{\Gamma}(m)/2$  and  $|\hat{g}'_\ell - \hat{g}_\ell| < \widehat{\Gamma}(m)/2$ , we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\widehat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\widehat{\Gamma}(m)}{2}) \geq \widehat{\Gamma}(m) - \widehat{\Gamma}(m) = 0.$$

Thus no coordinate in  $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$  can become smaller than any coordinate in  $\widehat{\mathcal{M}}^{\text{mean}}(g)$  under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since  $\widehat{\Gamma}(m)$  is fixed under conditioning on  $X$ , plugging  $\varepsilon = \widehat{\Gamma}(m)/2$  into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

### C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator on each input. Thus, arbitrarily small perturbations to outputs can change decisions with non-vanishing probability. We still consider the same stochastic perturbation model  $g'(x) = g(x) + \eta(x)$ .

**Lemma C.2** (Instability of per-sample argmin voting). *For a fixed input  $x$ , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

*denote the argmin and its margin. Then, for any fixed  $x$ ,*

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

*and for  $n$  i.i.d. inputs  $x_1, \dots, x_n$ ,*

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

*where  $\alpha := \mathbb{E}_x \left[ \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$ .*

**Interpretation.** Unlike mean extraction, voting does not benefit from variance reduction: each input incurs a direct hard decision based on noisy outputs. Increasing  $n$  only increases the number of opportunities for index-flips.

*Proof.* We first bound the per-input change probability  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$ . Fix  $x$  and write  $j^* := j^*(x; g)$  for brevity. By definition of the margin,  $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$ . Let  $j^{(2)}$  be an index attaining this minimum, i.e.,  $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$ .

Under the perturbation  $g'(x) = g(x) + \eta(x)$ , we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since  $\eta_{j^{(2)}}(x)$  and  $\eta_{j^*}(x)$  are independent  $\mathcal{N}(0, \sigma^2)$  variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes,  $j^*(x; g') \neq j^*(x; g)$ , certainly occurs whenever  $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$ . Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let  $Z \sim \mathcal{N}(0, 1)$ . Since  $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$ , we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore,  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$ .

Now consider  $n$  i.i.d. inputs  $x_1, \dots, x_n$ . Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across  $i$ , because the perturbations  $\eta(x_i)$  are independent across  $i$ . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define  $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$ , and  $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$ . Taking expectations over the randomness of  $x_i$ , we obtain  $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$ . Therefore, by independence across  $i$ ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function  $u \mapsto \log(1 - u)$  on  $u \in [0, 1)$ . Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set  $\widehat{\mathcal{M}}^{\text{vote}}$  is determined by the empirical counts of the dimensions  $\{j^*(x_i; g)\}_{i=1}^n$ . If the empirical gap between the  $m$ -th and  $(m+1)$ -th most frequent dimensions is of order  $O(1/n)$  (i.e., only a constant number of votes), then a change in  $j^*(x_i)$  for a single input  $x_i$  is sufficient to swap the ordering across this boundary and thus alter the top- $m$  set. Combining this observation with the lower bound  $1 - (1 - \alpha)^n$  shows that, in such regimes, the probability that  $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$  is bounded away from zero uniformly in  $n$ , i.e., per-sample argmin voting does not enjoy variance reduction with more samples.  $\square$

## C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze how many random inputs are required for the empirical mean estimator to recover the true bias. We reuse notation from **Setup**: each output coordinate  $g_j(x)$  is a random variable under random inputs  $x \sim \mathcal{D}$ , with population mean  $\bar{g}_j := \mathbb{E}[g_j(x)]$ .  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  represents the sorted means in non-decreasing order, and  $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$  is the separation margin. We estimate  $\bar{g}$  using  $n$  i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is  $\mathcal{M}$  with size  $m$  and our empirical estimation gives  $\widehat{\mathcal{M}}$ .

**Lemma C.3** (Sufficient condition for recovering the true set). *If  $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$  and  $2\varepsilon < \gamma$ , then the empirical bottom- $m$  set equals the true set:  $\widehat{\mathcal{M}} = \mathcal{M}$ .*

*Proof.* For any  $j \in \mathcal{M}$  and  $j' \notin \mathcal{M}$ , we have  $\bar{g}_j \leq \bar{g}_{(m)}$  and  $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$ . The  $\ell_\infty$  bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- $m$  set.  $\square$

We next interpret this recovery condition in terms of hyperparameters. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is  $\sigma$ -sub-Gaussian with  $\sigma$  proportional to its range. In our setting, each  $g_j(x)$  corresponds to a logit or final hidden activation, which are typically bounded or light-tailed in both initialized networks (due to controlled parameter initialization) and trained networks.

**Corollary C.4** (Sample complexity under sub-Gaussian coordinates). *Suppose each  $g_j(x)$  is  $\sigma$ -sub-Gaussian. Then with probability at least  $1 - \delta$ ,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .

*Proof.* For fixed  $j$ , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to  $\delta$  and solve for  $\varepsilon$  to obtain, with probability at least  $1 - \delta$ ,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If  $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$  (i.e., if  $2\varepsilon < \gamma$ ), then Lemma C.3 applies and implies  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .  $\square$

Table 9: Birth-level fingerprint separation under different random initializations across model families.

(a) LLaMA-style models: different seeds produce distinct SeedPrints

| Seed Pair                               | Logits Output  |                | Hidden State   |                |
|-----------------------------------------|----------------|----------------|----------------|----------------|
|                                         | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{42}^{init}$ vs. $s_{2000}^{base}$   | 0.404          | 0.456          | 0.357          | 0.532          |
| $s_{123}^{init}$ vs. $s_{42}^{base}$    | 0.214          | 0.295          | 0.678          | 0.565          |
| $s_{1000}^{init}$ vs. $s_{123}^{base}$  | 0.219          | 0.246          | 0.363          | 0.335          |
| $s_{2000}^{init}$ vs. $s_{1000}^{base}$ | 0.282          | 0.291          | 0.434          | 0.481          |

(b) Qwen-style models: different seeds produce distinct SeedPrints

| Seed Pair                              | Logits Output  |                | Hidden State   |                |
|----------------------------------------|----------------|----------------|----------------|----------------|
|                                        | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{123}^{init}$ vs. $s_{1000}^{base}$ | 0.741          | 0.727          | 0.094          | 0.074          |
| $s_{1000}^{init}$ vs. $s_{123}^{base}$ | 0.954          | 0.971          | 0.125          | 0.094          |
| $s_{42}^{init}$ vs. $s_{2000}^{base}$  | 0.273          | 0.360          | 0.451          | 0.529          |
| $s_{2000}^{init}$ vs. $s_{42}^{base}$  | 0.215          | 0.206          | 0.230          | 0.295          |

Table 10: (LLaMA-style models) Training preserves each model’s initialization fingerprint ( $p$ -value < 0.01).

| Model Pair                              | Logits Output  |                | Hidden State   |                | Baselines |        |        |        |
|-----------------------------------------|----------------|----------------|----------------|----------------|-----------|--------|--------|--------|
|                                         | <i>t</i> -test | <i>u</i> -test | <i>t</i> -test | <i>u</i> -test | Intrinsic | REEF   | PCS    | ICS    |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 3.33e-3✓       | 1.02e-3✓       | 2.20e-8✓       | 6.28e-8✓       | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.06e-3✓       | 7.33e-3✓       | 7.09e-6✓       | 1.37e-5✓       | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 2.44e-3✓       | 4.14e-3✓       | 5.58e-4✓       | 2.81e-3✓       | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 5.63e-3✓       | 6.76e-3✓       | 4.00e-10✓      | 1.27e-9✓       | -0.337×   | 0.331× | 0.581× | 0.188× |

Therefore, accurate recovery requires  $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$ , meaning the number of random queries grows logarithmically with the output dimension, but scales inversely with the squared separation margin  $\gamma$ . The margin  $\gamma$  itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of  $m$ .** Different choices of  $m$  correspond to different parts of the ranked spectrum  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$ , thus yielding different effective margins in practice.

The former can be controlled by the input length  $l$ , while the latter is selected empirically based on the observed gaps in the mean spectrum.

## D MORE EXPERIMENT DETAILS

### D.1 IMPLEMENTATION DETAILS AND LICENSES

For reproducibility, all models are trained with the Hugging Face Transformers Trainer (Wolf et al., 2020) and Accelerate (Gugger et al., 2022) for distributed runs. Open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License, and other models under Apache-2.0. Datasets are downloaded through the Hugging Face Datasets library, which is Apache-2.0, while each dataset’s content follows its stated license.

### D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This appendix provides complementary evidence for the experimental claims in Section 5. Instead of the analytical null used in the main text, we report results from the empirical simulation-based null in Algorithm 1. We evaluate both *logits* and *hidden state* outputs, and we apply both the one-sided *t*-test (*t*-test) and Mann–Whitney *U* test (*U* test).

#### D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

These complementary experiments test the same birth-to-lifecycle claims under a stochastic empirical null. We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al.,

Table 11: (Qwen-style models) Training preserves each model’s initialization fingerprint ( $p$ -value  $< 0.01$ ).

| Model Pair                              | Logits Output |           | Hidden State |           |
|-----------------------------------------|---------------|-----------|--------------|-----------|
|                                         | $t$ -test     | $U$ -test | $t$ -test    | $U$ -test |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.38e-03      | 1.68e-03  | 7.36e-15     | 3.38e-13  |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 1.35e-04      | 4.84e-05  | 4.41e-13     | 2.01e-11  |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 1.39e-03      | 1.46e-03  | 1.06e-24     | 2.05e-19  |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 1.80e-03      | 1.28e-03  | 4.87e-24     | 1.92e-20  |

Table 12: Complementary evidence for SeedPrints persistence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext).  $U$ -test refers to the Mann–Whitney  $U$  test.

(a) LLaMA-style models: simulation-null lineage verification results

| Setting                 | Ours (logits)      |                        | Ours (hidden)      |                       | Baselines          |                    |                    |                    |
|-------------------------|--------------------|------------------------|--------------------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Continual corpus (seed) | $t$ -test          | $U$ -test              | $t$ -test          | $U$ -test             | Intrinsic          | REEF               | PCS                | ICS                |
| TinyStories (1000)      | 0 <sup>✓</sup>     | 0 <sup>✓</sup>         | 0 <sup>✓</sup>     | 7.77e-89 <sup>✓</sup> | 1.000 <sup>✓</sup> | 0.759 <sup>×</sup> | 0.999 <sup>✓</sup> | 0.996 <sup>✓</sup> |
| TinyStories (123)       | 1.000 <sup>✓</sup> | 1.00 <sup>✓</sup>      | 0.943 <sup>✓</sup> | 0.902 <sup>✓</sup>    | 0.950 <sup>×</sup> | 0.658 <sup>✓</sup> | 0.332 <sup>✓</sup> | 0.012 <sup>✓</sup> |
| the_stack (1000)        | 0 <sup>✓</sup>     | 1.73e-287 <sup>✓</sup> | 0 <sup>✓</sup>     | 3.09e-69 <sup>✓</sup> | 0.489 <sup>×</sup> | 0.557 <sup>×</sup> | 0.585 <sup>×</sup> | 0.123 <sup>×</sup> |
| the_stack (123)         | 0.616 <sup>✓</sup> | 0.479 <sup>✓</sup>     | 0.732 <sup>✓</sup> | 0.831 <sup>✓</sup>    | 0.445 <sup>✓</sup> | 0.580 <sup>✓</sup> | 0.301 <sup>✓</sup> | 0.026 <sup>✓</sup> |

(b) Qwen-style models: simulation-null lineage verification results

| Setting                 | Ours (logits)         |                        | Ours (hidden)          |                       | Baselines          |                    |                    |                    |
|-------------------------|-----------------------|------------------------|------------------------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Continual corpus (seed) | $t$ -test             | $U$ -test              | $t$ -test              | $U$ -test             | Intrinsic          | REEF               | PCS                | ICS                |
| TinyStories (1000)      | 5.36e-09 <sup>✓</sup> | 1.92e-07 <sup>✓</sup>  | 8.49e-214 <sup>✓</sup> | 5.09e-71 <sup>✓</sup> | 1.000 <sup>✓</sup> | 0.957 <sup>✓</sup> | 0.999 <sup>✓</sup> | 0.996 <sup>✓</sup> |
| TinyStories (123)       | 0.434 <sup>✓</sup>    | 0.433 <sup>✓</sup>     | 0.256 <sup>✓</sup>     | 0.065 <sup>✓</sup>    | 0.913 <sup>×</sup> | 0.199 <sup>✓</sup> | 0.328 <sup>✓</sup> | 0.039 <sup>✓</sup> |
| the_stack (1000)        | 0 <sup>✓</sup>        | 4.16e-237 <sup>✓</sup> | 1.16e-211 <sup>✓</sup> | 2.30e-76 <sup>✓</sup> | 0.999 <sup>✓</sup> | 0.313 <sup>×</sup> | 0.995 <sup>✓</sup> | 0.976 <sup>✓</sup> |
| the_stack (123)         | 0.999 <sup>✓</sup>    | 0.993 <sup>✓</sup>     | 0.610 <sup>✓</sup>     | 0.491 <sup>✓</sup>    | 0.916 <sup>×</sup> | 0.255 <sup>✓</sup> | 0.328 <sup>✓</sup> | 0.038 <sup>✓</sup> |

2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we average  $p$ -values over 10 independent trials and use significance level  $\alpha = 0.01$ . Extremely small  $p$ -values should be interpreted only as decisive rejection: after values fall below numerical and sampling noise, for example  $< 10^{-20}$ , a value such as  $10^{-260}$  is not meaningfully stronger than  $10^{-20}$ .

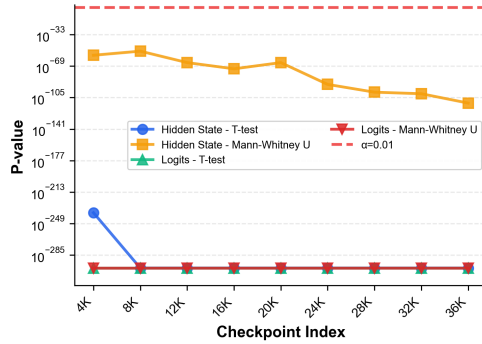
**Different initialization seeds produce distinct fingerprints.** Table 9a and Table 9b report  $p$ -values for different-seed model pairs (42, 123, 1000, and 2000) under both the  $t$ -test and  $U$ -test for LLaMA-style and Qwen-style models. All  $p$ -values remain  $> 0.01$ , so the empirical-null variant also separates models at birth rather than conflating distinct random seeds.

**Training preserves the initialization fingerprint.** Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model  $s_i^{init}$  with its OpenWebText-trained descendant  $s_i^{base}$  (Gokaslan et al., 2019) ( $\approx 10$ B tokens). Across all seed-model pairs,  $p$ -values are  $< 0.01$ , showing that the descendant retains the initialization bias profile. Baselines in Table 10 fail to distinguish across seeds, reinforcing the main-text conclusion that their signals are training-induced rather than initialization-born.

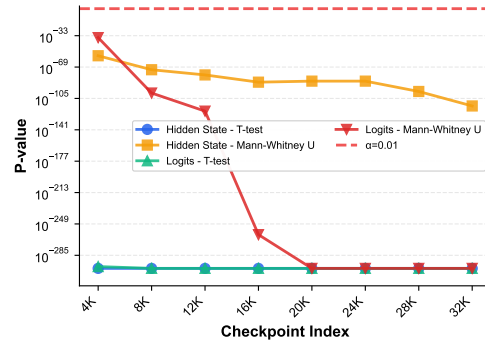
**Identical data and order do not make fingerprints converge.** Table 13a (LLaMA-style) and Table 13b (Qwen-style) isolate the role of initialization. All suspicious models  $s_i^{base}$  for  $i \in \{42, 123, 100, 2000\}$  use exactly the same OpenWebText corpus and data order; the training seed fixes the curriculum, leaving only the initialization seed  $i$  to vary. Cross-seed  $p$ -values stay  $> 0.01$ , unlike the near-zero same-seed values in Table 10. Thus, even with the empirical null, shared data and curriculum do not erase seed-specific fingerprints.

Table 13: Simulation-null check: identical dataset and training order do not make different initializations share SeedPrints.

| (a) LLaMA-style models                  |                |                |                |                | (b) Qwen-style models                  |                |                |                |                |
|-----------------------------------------|----------------|----------------|----------------|----------------|----------------------------------------|----------------|----------------|----------------|----------------|
| Model Pair                              | Logits Output  |                | Hidden State   |                | Model Pair                             | Logits Output  |                | Hidden State   |                |
|                                         | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |                                        | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{123}^{init}$ vs. $s_{1000}^{base}$  | 0.484          | 0.500          | 0.385          | 0.486          | $s_{123}^{init}$ vs. $s_{1000}^{base}$ | 0.598          | 0.638          | 0.286          | 0.254          |
| $s_{1000}^{init}$ vs. $s_{2000}^{base}$ | 0.946          | 0.956          | 0.135          | 0.096          | $s_{1000}^{init}$ vs. $s_{123}^{base}$ | 0.804          | 0.805          | 0.236          | 0.240          |
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.598          | 0.589          | 0.426          | 0.337          | $s_{42}^{init}$ vs. $s_{2000}^{base}$  | 0.589          | 0.608          | 0.226          | 0.2043         |
| $s_{2000}^{init}$ vs. $s_{42}^{base}$   | 0.756          | 0.781          | 0.388          | 0.287          | $s_{2000}^{init}$ vs. $s_{42}^{base}$  | 0.523          | 0.482          | 0.312          | 0.323          |



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: All-stage simulation-null verification: SeedPrints detects lineage at every checkpoint ( $p$ -values  $< 0.01$ ) across model families.

**Continual training on diverse datasets does not confound the fingerprint.** The copyright-relevant contribution is not merely seed separation; it is robustness to downstream data shifts that can mislead attribution. Prior methods may fail when continued training changes the domain because their signals track distributional similarity instead of lineage.

In Table 12a and Table 12b, we continue training a seed-1000 base pretrained on OpenWebText (Gokaslan et al., 2019) on TinyStories (Eldan & Li, 2023), a synthetic children’s-story corpus, and The Stack (Kocetkov et al., 2022), a permissively licensed code corpus. We compare true descendants against *distractors* derived from a seed-123 base trained with a different OpenWebText order and then continued on the same target corpus. The task is to identify which model truly shares lineage with the source, not which model shares the closest downstream domain.

The baselines fail in the code setting (The Stack), misclassifying true descendants as distractors, which indicates sensitivity to domain similarity rather than lineage identity. SeedPrints correctly attributes lineage across both corpora, supporting the claim that the initialization-born fingerprint survives substantial domain shift and persists beyond the initial pretraining stage.

#### D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** We finally verify the all-stage claim in a controlled OpenWebText trajectory with known initialization seed. Each intermediate checkpoint is treated as a source, and SeedPrints tests whether its offspring remains identifiable along the same training path. All variants recognize the suspect as same-lineage, with  $p$ -values below 0.01 for both LLaMA-style models (Figure 5a) and Qwen-style models (Figure 5b). This confirms that the empirical-null version remains stable under continuous optimization.

#### D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

This study checks an important boundary condition for the contribution: SeedPrints should not collapse distinct model sizes within the same family into a single family-level fingerprint. We

Table 14: Cross-size SeedPrints check within Qwen-2.5: same-family models are not automatically classified as sharing lineage with Qwen-2.5-14B.

|                          | Qwen-2.5-0.5B          | Qwen-2.5-1.5B          | Qwen-2.5-3B            | Qwen-2.5-7B            |
|--------------------------|------------------------|------------------------|------------------------|------------------------|
| SeedPrint ( $\uparrow$ ) | 0.7184                 | 0.9707                 | 0.3898                 | 0.5429                 |
| REEF ( $\downarrow$ )    | 0.8250                 | 0.8399                 | 0.8833                 | 0.8461                 |
| HUREF ( $\downarrow$ )   | $-6.37 \times 10^{-5}$ | $-2.84 \times 10^{-5}$ | $-8.70 \times 10^{-6}$ | $-6.08 \times 10^{-6}$ |

therefore compare Qwen-2.5-14B as the protected target with Qwen-2.5- $\{0.5B, 1.5B, 3B, 7B\}$  as suspects that share the same architectural family but are not descendants of Qwen-2.5-14B. Table 14 reports SeedPrint ( $\uparrow$ ) alongside REEF ( $\downarrow$ ) and HUREF ( $\downarrow$ ). As in the main paper, SeedPrints outputs a  $p$ -value for the hypothesis that the suspect shares initialization-born fingerprints with the target, whereas REEF and HUREF output continuous similarity scores that require a threshold.

The first observation supports the intended contribution. All SeedPrints  $p$ -values are above a typical significance level, such as  $\alpha=0.01$ , so none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This is the desired behavior: shared family design and similar architecture do not cause SeedPrints to conflate distinct pretraining runs.

The cross-size  $p$ -values also spread across the  $[0, 1]$  range rather than collapsing to one value. We do *not* treat these magnitudes as a continuous similarity score; their role is to support a calibrated yes/no decision at a fixed significance level. Still, the spread suggests that the test statistic remains sensitive to the concrete combination of architecture and initialization seed instead of becoming a generic Qwen-2.5 family signature.

The similarity-based baselines show why a statistical decision rule is useful. REEF produces a narrow score band (0.8250–0.8833) across sizes, while HUREF gives values close to zero (on the order of  $10^{-5}$ ). Because neither method provides a built-in decision rule, users must choose thresholds manually, and a single cutoff may not cleanly separate cross-size variants from true descendants. SeedPrints instead gives a decision at a prescribed significance level, directly answering whether two same-family models plausibly share training lineage.

## E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we also indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” refer to *pre-trained* and *instruction-tuned* source models, respectively.

| Parameter        | Value |
|------------------|-------|
| max_new_tokens   | 512   |
| temperature      | 0.7   |
| top_p            | 0.9   |
| top_k            | 50    |
| do_sample        | true  |
| max_input_length | 512   |

## E.1 MODEL LIST BY FAMILY

### E.1.1 QWEN-2.5-7B

| Variant | HuggingFace repo                             | Type         |
|---------|----------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-7B                              | —            |
| IT      | Qwen/Qwen2.5-7B-Instruct                     | —            |
| Derived | Qwen/Qwen2.5-Math-7B                         | finetune     |
| Derived | Qwen/Qwen2.5-Coder-7B-Instruct               | finetune     |
| Derived | WangCa/Qwen2.5-7B-Medicine                   | finetune     |
| Derived | huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2 | finetune     |
| Derived | Locutusque/StockQwen-2.5-7B                  | merge        |
| Derived | bunnycore/QevaCoT-7B-Stock                   | merge        |
| Derived | fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct | adapter      |
| Derived | SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct    | adapter      |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4           | quantization |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8           | quantization |
| Derived | Lansechen/Qwen2.5-7B-Open-R1-Distill         | distillation |

### E.1.2 QWEN2.5-14B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-14B                                | —            |
| IT      | Qwen/Qwen2.5-14B-Instruct                       | —            |
| Derived | Qwen/Qwen2.5-Coder-14B                          | finetune     |
| Derived | oxyapi/oxy-1-small                              | finetune     |
| Derived | v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno | merge        |
| Derived | ToastyPigeon/qwen-story-test-qlora              | adapter      |
| Derived | Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4             | quantization |
| Derived | deepseek-ai/DeepSeek-R1-Distill-Qwen-14B        | distillation |

### E.1.3 LLAMA-3.1-8B

| Variant | HuggingFace repo                                                 | Type         |
|---------|------------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-3.1-8B                                          | —            |
| IT      | meta-llama/Llama-3.1-8B-Instruct                                 | —            |
| Derived | ValiantLabs/Llama3.1-8B-Fireplace2                               | finetune     |
| Derived | RedHatAI/Llama-3.1-8B-tldr                                       | finetune     |
| Derived | proxectonos/Llama-3.1-Carballo                                   | finetune     |
| Derived | mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated                  | finetune     |
| Derived | gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES       | merge        |
| Derived | XiaoJian9992024/Llama3.1-8B-ExtraMix                             | merge        |
| Derived | LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching | adapter      |
| Derived | chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7                     | adapter      |
| Derived | iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8                   | quantization |
| Derived | DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ                            | quantization |
| Derived | asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill                    | distillation |

#### E.1.4 MISTRAL-7B-v0.3

| Variant | HuggingFace repo                                    | Type         |
|---------|-----------------------------------------------------|--------------|
| PT      | mistralai/Mistral-7B-v0.3                           | —            |
| IT      | mistralai/Mistral-7B-Instruct-v0.3                  | —            |
| Derived | KurmaAI/AQUA-7B                                     | finetune     |
| Derived | openfoodfacts/spellcheck-mistral-7b                 | finetune     |
| Derived | grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B      | merge        |
| Derived | chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer     | adapter      |
| Derived | RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit         | quantization |
| Derived | eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b | distillation |

#### E.1.5 GEMMA-2-2B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | google/gemma-2-2b                               | —            |
| IT      | google/gemma-2-2b-it                            | —            |
| Derived | rinna/gemma-2-baku-2b                           | finetune     |
| Derived | anakin87/gemma-2-2b-neogenesis-ita              | finetune     |
| Derived | vonjack/gemma2-2b-merged                        | merge        |
| Derived | google-cloud-partnership/gemma-2-2b-it-lora-sql | adapter      |
| Derived | qilowoq/gemma-2-2B-it-4Bit-GPTQ                 | quantization |
| Derived | Syed-Hasan-8503/Gemma-2-2b-it-distilled         | distillation |

#### E.1.6 LLAMA-2-7B

| Variant | HuggingFace repo                                                | Type         |
|---------|-----------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-2-7b-hf                                        | —            |
| IT      | meta-llama/Llama-2-7b-chat-hf                                   | —            |
| Derived | allenai/tulu-2-7b                                               | finetune     |
| Derived | QIAIUNCC/EYE-Llama_qa                                           | finetune     |
| Derived | DevQuasar/coma-7B-v0.1                                          | merge        |
| Derived | Ammar-1/llama2-Better-Tune                                      | adapter      |
| Derived | TheBloke/Llama-2-7B-Chat-GPTQ                                   | quantization |
| Derived | cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | distillation |

#### E.1.7 TINYLLAMA-1.1B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT/IT   | TinyLlama/TinyLlama-1.1B-Chat-v1.0              | —            |
| Derived | alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2 | finetune     |
| Derived | Edentns/DataVortexTL-1.1B-v0.1                  | finetune     |
| Derived | appvoid/dot-v2.7                                | merge        |
| Derived | barissglc/tinyllama-tarot-v1                    | adapter      |
| Derived | TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ          | quantization |
| Derived | anudaw/distilled-code-llama                     | distillation |

**Notes.** “Type” denotes the post-training transformation relative to the PT/IT source model: *finetune* includes supervised/preference optimization variants; *adapter* includes LoRA/QLoRA-style modules; *merge* includes model soups and TIES-style merges; *quantization* includes GPTQ/INTx variants; *distillation* includes student models distilled from reasoning-augmented teachers.

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## E.2 SAMPLE RESULTS

We include partial results from one LeafBench evaluation run in Table 15 using the  $u$ -test with the simulation-based null. The table reports  $1 - p$  as a benchmark-compatible lineage score. We use a significance level of 0.01 for  $p$ -values; empirically, LeafBench selects an optimal decision threshold of 0.9920, which is consistent with strong separation in this run.

The important interpretation is distributional rather than linear. Individual  $p$ -values should not be read as similarity magnitudes, but across model pairs they reveal the fingerprint structure: same-lineage pairs produce much lower  $p$ -values, while unrelated pairs remain close to uniform.

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| Model                                                      | Qwen-2.5-7B | Qwen-2.5-7B-Instruct | Qwen2.5-14B | Qwen2.5-14B-Instruct |
|------------------------------------------------------------|-------------|----------------------|-------------|----------------------|
| Qwen-2.5-7B                                                | 1.000       | 1.000                | 0.079       | 0.000                |
| Qwen-2.5-7B-Instruct                                       | 1.000       | 1.000                | 0.813       | 0.536                |
| Qwen2.5-7B-Math                                            | 1.000       | 0.994                | 0.047       | 0.987                |
| Qwen2.5-7B-Coder                                           | 0.996       | 0.999                | 0.177       | 0.002                |
| Qwen2.5-7B-Instruct-Medicine                               | 1.000       | 1.000                | 0.576       | 0.369                |
| Qwen2.5-7B-Instruct-Abiliterated                           | 1.000       | 1.000                | 0.276       | 0.886                |
| Qwen2.5-7B-Stock                                           | 1.000       | 1.000                | 0.939       | 0.225                |
| QevaCoT-7B                                                 | 1.000       | 1.000                | 0.199       | 0.119                |
| Qwen2.5-7B-Instruct-Task-10                                | 1.000       | 1.000                | 0.672       | 0.090                |
| Qwen2.5-7B-Instruct-Task-12                                | 1.000       | 1.000                | 0.155       | 0.987                |
| Qwen2.5-7B-Instruct-Int4                                   | 1.000       | 1.000                | 0.827       | 0.961                |
| Qwen2.5-7B-Instruct-Int8                                   | 1.000       | 1.000                | 0.914       | 0.739                |
| Qwen2.5-7B-Open-R1-Distill                                 | 1.000       | 1.000                | 0.302       | 0.955                |
| Qwen2.5-14B                                                | 0.324       | 0.487                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct                                       | 0.403       | 0.702                | 1.000       | 1.000                |
| Qwen2.5-Coder-14B                                          | 0.492       | 0.660                | 1.000       | 1.000                |
| oxy-1-small                                                | 0.492       | 0.104                | 1.000       | 1.000                |
| Qwen2.5-14B-Gutenberg-Instruct-Slerpeno                    | 0.856       | 0.931                | 1.000       | 1.000                |
| Qwen-story-test-qlora                                      | 0.112       | 0.898                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct-GPTQ-Int4                             | 0.583       | 0.092                | 1.000       | 1.000                |
| DeepSeek-R1-Distill-Qwen-14B                               | 0.204       | 0.000                | 1.000       | 1.000                |
| Llama-3.1-8B                                               | 0.638       | 0.714                | 0.156       | 0.000                |
| Llama-3.1-8B-Instruct                                      | 0.643       | 0.118                | 0.128       | 0.510                |
| Llama-3.1-8B-Fireplace2                                    | 0.715       | 0.532                | 0.065       | 0.354                |
| Llama-3.1-8B-TLDR                                          | 0.925       | 0.811                | 0.331       | 0.014                |
| Llama-3.1-8B-Carballo                                      | 0.001       | 0.605                | 0.003       | 0.946                |
| Llama-3.1-8B-Instruct-Abliterated                          | 0.061       | 0.005                | 0.121       | 0.422                |
| Llama-3.1-8B-Instruct-HalfAbliterated-TIES                 | 0.356       | 0.509                | 0.796       | 0.000                |
| Llama-3.1-8B-ExtraMix                                      | 0.640       | 0.160                | 0.721       | 0.875                |
| Llama-3.1-8B-Instruct-cv-job-description-matching          | 0.684       | 0.292                | 0.004       | 0.001                |
| Llama-3.1-8B-Instruct-PsyCourse-fold7                      | 0.682       | 0.057                | 0.063       | 0.191                |
| Llama-3.1-8B-Instruct-8bit                                 | 0.798       | 0.214                | 0.154       | 0.385                |
| Llama-3.1-8B-Instruct-4bit                                 | 0.513       | 0.023                | 0.005       | 0.933                |
| Llama-3.1-8B-Instruct-Open-R1-Distill                      | 0.800       | 0.101                | 0.772       | 0.024                |
| Mistral-7B-v0.3                                            | 0.224       | 0.352                | 0.506       | 0.044                |
| Mistral-7B-v0.3-Instruct                                   | 0.000       | 0.000                | 0.984       | 0.104                |
| AQUA-7B                                                    | 0.014       | 0.613                | 0.703       | 0.569                |
| Mistral-7B-v0.3-Spellcheck                                 | 0.005       | 0.481                | 0.872       | 0.105                |
| Mistral-7B-v0.3-Instruct-demi-merge                        | 0.904       | 0.774                | 0.058       | 0.522                |
| Mistral-7B-v0.3-Brain                                      | 0.340       | 0.983                | 0.481       | 0.016                |
| Mistral-7B-v0.3-Instruct-GPTQ-4bit                         | 0.002       | 0.281                | 0.089       | 0.784                |
| Mistral-7B-distilled-from-deepseek-r1-qwen32b              | 0.009       | 0.001                | 0.511       | 0.015                |
| Gemma-2-2b                                                 | 0.747       | 0.869                | 0.689       | 0.000                |
| Gemma-2-2b-it                                              | 0.944       | 0.039                | 0.626       | 0.060                |
| Gemma-2-baku-2b                                            | 0.616       | 0.231                | 0.176       | 0.464                |
| Gemma-2-2b-neogenesis-ita                                  | 0.002       | 0.536                | 0.079       | 0.980                |
| Gemma-2-2b-merged                                          | 0.951       | 0.012                | 0.651       | 0.075                |
| Gemma-2-2b-it-lora-sql                                     | 0.236       | 0.408                | 0.000       | 0.000                |
| Gemma-2-2B-it-4Bit-GPTQ                                    | 0.945       | 0.159                | 0.000       | 0.008                |
| Gemma-2-2b-it-distilled                                    | 0.482       | 0.573                | 0.271       | 0.276                |
| Llama-2-7b                                                 | 0.443       | 0.470                | 0.148       | 0.276                |
| Llama-2-7b-chat                                            | 0.528       | 0.868                | 0.424       | 0.766                |
| tulu-2-7b                                                  | 0.125       | 0.272                | 0.170       | 0.888                |
| EYE-Llama_qa                                               | 0.657       | 0.244                | 0.674       | 0.968                |
| coma-7B-v0.1                                               | 0.953       | 0.004                | 0.573       | 0.731                |
| llama2-Better-Tune                                         | 0.136       | 0.699                | 0.469       | 0.860                |
| Llama-2-7B-Chat-GPTQ                                       | 0.594       | 0.603                | 0.582       | 0.424                |
| llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | 0.349       | 0.224                | 0.366       | 0.976                |

Table 15: **Sample SeedPrints results.** Pairwise lineage scores between target models (rows) and base models (columns). Cells with  $1 - p \geq 0.99$  are highlighted in green as same-lineage pairs.